

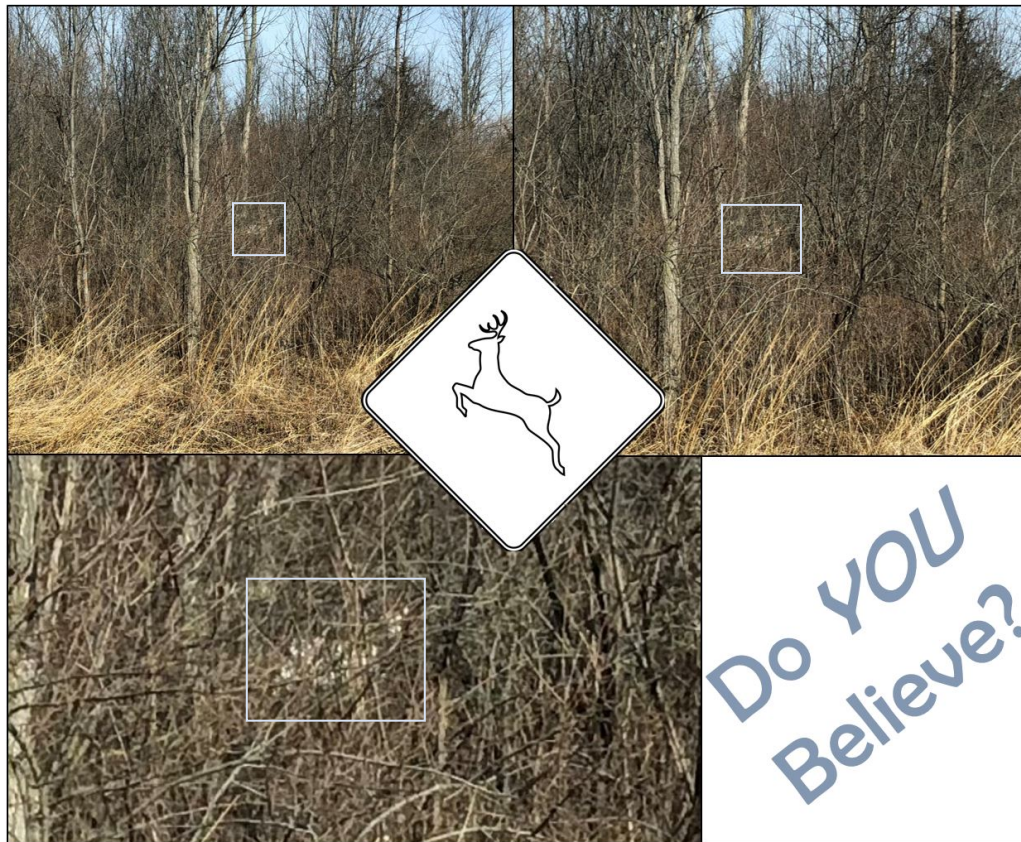
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BIOL 325 – Invasion Ecology

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Post Card #2



3. The phylogenetic origin of species and how this relates to invasibility.

The North American White-tailed Deer, *Odocoileus virginianus*, is a well-known species of herbivorous artiodactyl mammals. They typically reside in forested areas of North America, expressing a dominant fur color of brown. At the Seneca Army Depot in Geneva, NY, a population of *O. virginianus* live within the fenced-in property. Back in 1949, the Army General noticed that one of the female deer gave birth to two, white-furred fawns. This white fur coloration is indicative of recessive gene expression. Regardless of differing phenotypic expression, the dominant brown-colored deer and the recessive white-colored deer are still considered the same species taxonomically. The Army General was so impressed by these unusually colored deer that he declared them protected, banning any hunting of them. By limiting the act of human predation on these white deer, artificial selection favored the trait coding for white fur. Through this process of artificial selection, the white deer were able to

establish and maintain population levels well enough to now be considered the largest herd of *white*, white-tailed deer in North America.